

Gauss Law

CHAPTER 23, R.H.W-10th Edition

Gauss' law In the chapter Electric field, we did a

labour-intensive job $\rightarrow$ We split the charge distribution up into charge element dq , found the field $d\vec{E}$ due to an element, and resolved the vector into components. Then we determined whether the components from all elements would end up cancelling or adding.

$\rightarrow$ One of the primary goal of science to find simple way to solve a problem. Symmetry is one of the such tool to solve problem.

■ In this chapter we will discuss a beautiful relationship between charge and electric field that allow us, in certain symmetric situations, to find the electric field of an extended charged object in simple way (few algebra). The relationship is called Gauss' law.

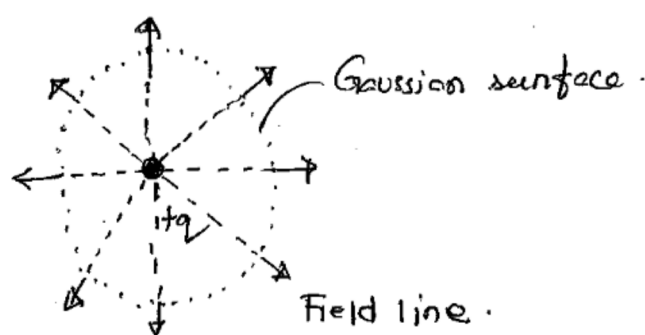
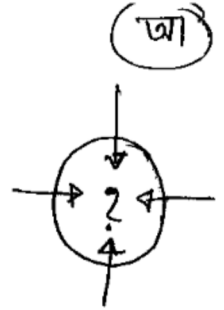
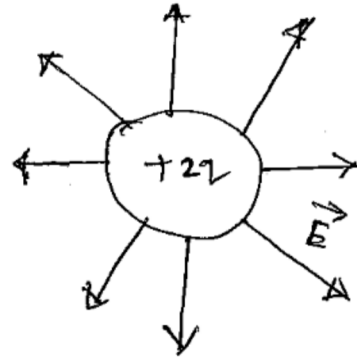
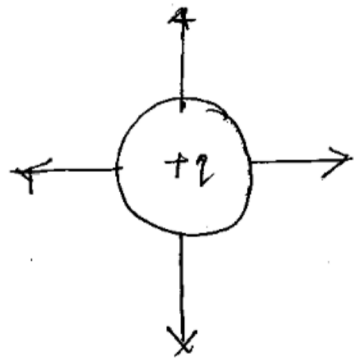


Figure shows a particle with charge $+Q$ that is surrounded by an imaginary concentric sphere. The surface of the sphere is said to be a homocentric Gaussian surface. At points on the sphere the electric field vectors have a moderate magnitied and point radially away from the particle. (positively charged).



Notation of Gauss's law

① If charge shift from $+q$ to $+2q$,

Some surface has

1. twice field vectors
2. density of the field lines would be twice.

□ Gauss' Law relates the electric field at points on a Gaussian surface to the net charge enclosed by that surface.

The problems in this chapter includes

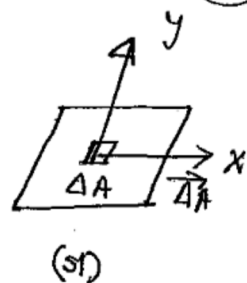
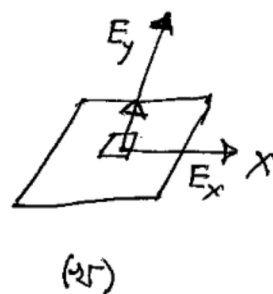
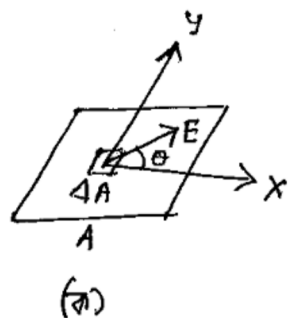
(i) We use Gauss' Law to find the field at some point.

(ii) Sometimes, we know the field on a Gaussian surface and we use Gauss' Law to find the charge enclosed by the surface.

→ So, we need a quantitative way of determining how much electric field pierces a surface.

→ The measured way is called electric flux.

Electric Flux



① Flat surface with area A in a uniform electric field $\vec{E}$.

② One of the piercing electric field $\vec{E}$ from area ΔA (means small).

③ Only x component is coming out of the surface and y component align or merged with the surface and does not come into play in Gauss' law.

The amount of electric field piercing the patch is defined to be the electric flux $\Delta\phi$ through it.

$$\Delta\phi = (E \cos\theta) \Delta A$$

$\Delta\phi = \vec{E} \cdot \vec{\Delta A}$. To find the total flux ϕ through the surface, we sum the flux through every patch on the surface: $\phi = \sum \vec{E} \cdot \vec{\Delta A}$

However we don't want to sum up 100's of these values, rather than, we transform the summation into an integral by shrinking the patches from small squares with area ΔA with area dA .

The total flux is then $\phi = \int \vec{E} \cdot d\vec{A}$ (total flux).

We can evaluate dot product in magnitude-angle notation: $E \cos\theta dA$. When the electric field is uniform and the surface is flat $E \cos\theta = \text{constant}$.

$\phi = (E \cos\theta) A$ (Uniform field, flat surface).

Closed Surface To use Gauss' law to relate flux and charge, we need a closed surface. Now we are interested not only the piercing components of the field but also on whether the piercing is inward or outward.

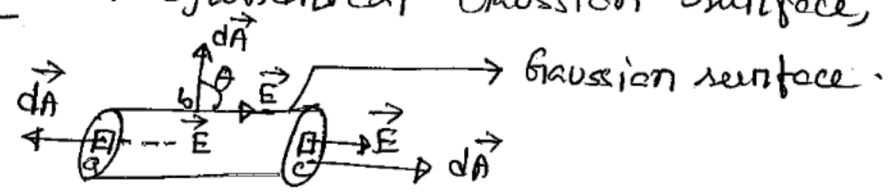
$\vec{E} \cdot \vec{dA}$, now θ is either 0 , or 180°

An inward piercing field is negative flux. An outward piercing field is positive flux. A skimming field is zero flux.

net flux $\phi = \oint \vec{E} \cdot d\vec{A}$ (net flux) — (1)

Sample problem 23.01

A cylindrical Gaussian surface,



Closed by end caps, is immersed in a uniform electric field. The cylinder axis is parallel to the field direction.

Cylinder radius R . It lies in a uniform electric field $\vec{E}$ with the cylinder's central axis, parallel to the field.

$$\begin{aligned} \phi &= \oint \vec{E} \cdot d\vec{A} \\ &= \int_a \vec{E} \cdot d\vec{A} + \int_b \vec{E} \cdot d\vec{A} + \int_c \vec{E} \cdot d\vec{A} \end{aligned} \quad \text{--- (2)}$$

$$\int_a \vec{E} \cdot d\vec{A} = \int_a E (\cos 180^\circ) dA = -E \int_a dA = -EA,$$

$$\int_c \vec{E} \cdot d\vec{A} = \int_c E (\cos 0^\circ) dA = EA$$

$$\int_b \vec{E} \cdot d\vec{A} = \int_b E (\cos 90^\circ) dA = 0$$

$$\phi = -EA + 0 + EA = 0$$

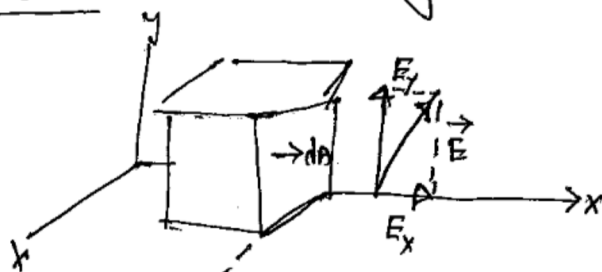
The net flux is zero, because the field lines that represent the electric field all pass entirely through the Gaussian surface.

Sample problem 23.02

Flux through a closed cube,



nonuniform field.



$$\vec{E} = 3.0x\hat{i} + 4.0\hat{j}$$

Q → What is the electric flux through the right face, the left face, and the top face?

We can find the flux Φ through the surface by integrating $(\vec{E} \cdot d\vec{A})$

Right face:

Area vector $d\vec{A}$ for any patch element on the right face of the cube must point in the positive direction of the x axis.

$$dA = dA\hat{i}$$

$$\begin{aligned}\Phi_r &= \int \vec{E} \cdot d\vec{A} = \int (3.0x\hat{i} + 4.0\hat{j}) \cdot dA\hat{i} = \int (3.0x dA + 0) \\ &= 3.0 \int x dA = 3.0 \times 3.0 \int dA = \frac{9}{C} \times 4.0 \text{ m}^2 \\ &= 36 \text{ N} \cdot \text{m}^2/\text{C}.\end{aligned}$$

Left face

$$x = 1.0 \text{ m}$$

$$\Phi_l = -12 \text{ N} \cdot \text{m}^2/\text{C}$$

Top face:

$$\begin{aligned}\Phi_t &= \int (3.0x\hat{i} + 4.0\hat{j}) \cdot (dA\hat{j}) \\ &= \int [(3.0x)(dA)\hat{i} \cdot \hat{j} + (4.0)(dA)\hat{j} \cdot \hat{j}] \\ &= \int 4 dA = 4 \int dA \\ &= 4 \text{ N} \cdot \text{m}^2 \times 4 \text{ m}^2 \\ &= 16 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

Gauss' Law

Gauss law relates the net flux Φ of an electric field through a closed surface (a Gaussian surface) to the net charge q_{enc} that is enclosed by the surface.

It tells us that,

$$\epsilon_0 \Phi = q_{enc} \text{ (Gauss Law).}$$

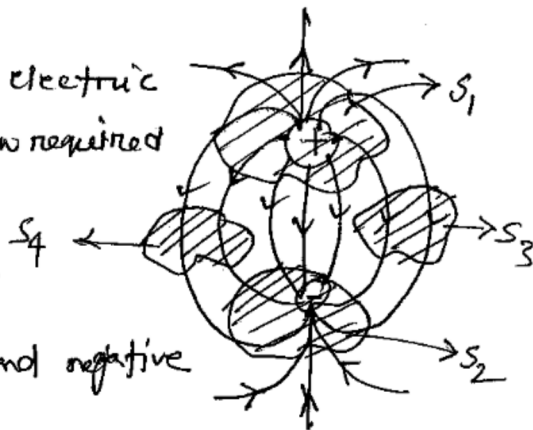
$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{enc} \text{ (Gauss Law).} \quad \text{---(1)}$$

$q_{enc} \rightarrow$ is the algebraic sum of all the enclosed positive and negative charges (It can be positive, negative and or zero charges). The sign tells us about the net flux through the Gaussian surface.

Note: The electric field due to a charge outside the Gaussian surface contributes zero net flux through the surface (Because as many field lines due to that charged enter the surface as leave it).

S_1 net charge is q_{enc} and electric field is outward. Gauss law required

S_2 net charge is q_{enc} and electric field is inward and negative



S_3 and S_4 Surface enclosed no net charge, enclosed positive and negative charges have equal magnitude.

Gauss Law and Coulomb's Law

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \text{[Coulomb's Law]}$$

for a charged particle q .

We can calculate the same electric field using Gauss' Law with symmetry. (Spherical symmetry)

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{enc.}$$

The direction of $d\vec{A}$ is radially outward, $\theta = 0$

$$\epsilon_0 \oint E dA = q_{enc} \quad | \quad q_{enc} = q$$

The field depends on the distance r from the particle, not the direction

$$\epsilon_0 E \oint dA = q$$

$$\epsilon_0 E (4\pi r^2) = q$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Symmetry in Gauss Law, is important

The field magnitude E is the same at every patch element, E can be pulled outside the integral

Sample Problem 23.03

Answer:

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = \epsilon_0 \oint E \cos\theta dA = \epsilon_0 E \oint dA$$

E is constant over the Gaussian surface.

$$E = \frac{q_{enc}}{4\pi\epsilon_0 r^2} = \frac{q_{enc}}{4\pi\epsilon_0 r^2}$$

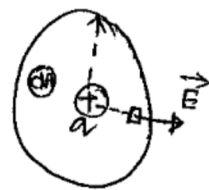
We can draw a surface (sphere)

$$q_{enc} = q + Q = 5e + (-16e) = -11e$$

$$\theta = \vec{E} \text{ and } d\vec{A} \rightarrow 180^\circ$$

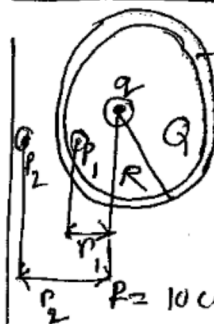
$$dA = -E dA \quad | \quad E = \frac{-q_{enc}}{4\pi\epsilon_0 r^2}$$

Gaussian surface.



A spherical Gaussian surface centered on a particle with charge q .

Problem illustration



Plastic spherical shell.

$Q = -16e$, spherical shell.

$q = +5e$ is at the center.

What is the electric field (magnitude and direction) at

(a) point P_1 at radial distance $r_1 = 6.0 \text{ cm}$. and

(b) point P_2 at radial distance $r_2 = 12.0 \text{ cm}$.

A Charged isolated Conductor

(19)

If an excess charge is placed on an isolated conductor, that amount of charge will move entirely to the surface of the conductor. None of the excess charge will be found within the body of the conductor.

Excess charge not inside the conductor but eventually stay on the surface of the conductor.

An Isolated Conductor with a Cavity

$\vec{E} = 0$ inside the conductor, there can be no flux through this new Gaussian surface.

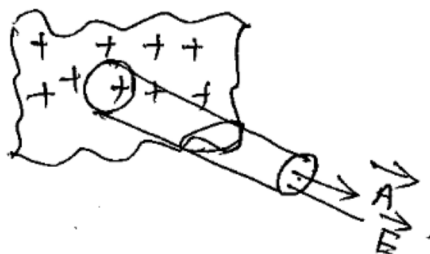
The External Electric field

○ The excess charge on an isolated conductor moves entirely to the conductor surface. Unless the conductor is spherical, the charge does not distribute itself uniformly.

→ The surface charge density σ varies over the surface of any nonspherical conductor.

— using Gauss law: Consider a surface that is considerably small enough to avoid any curvature.

$$q_{\text{enc}} = \sigma A$$



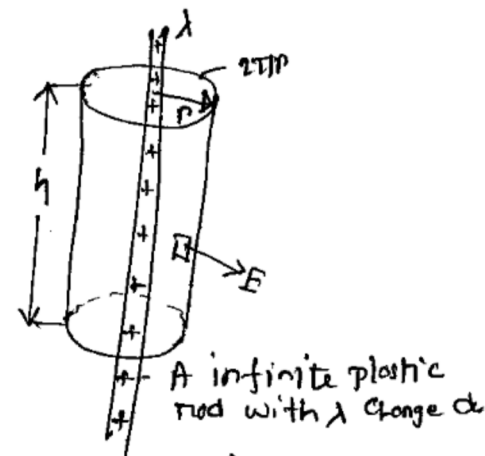
$$\epsilon_0 E A = \sigma A$$

$$E = \frac{\sigma}{\epsilon_0} \quad [\text{conducting surface}] \quad \boxed{E \propto \sigma}$$

Thus the magnitude of the electric field just outside a conductor is proportional to the surface charge density on the conductor. The sign of the charge gives the direction of the field.

Cylindrical Symmetry

We want to find an expression for the electric field magnitude E at radius r from the central axis of the rod, (outside the rod).



There is flux only through the curved surface.

→ Area vector $d\vec{A}$ is radially outward (away from the interior of the Gaussian surface). The dot product in Gauss' law is then simply $E dA \cos \theta = E dA$.

Gauss Eqⁿ becomes $\epsilon_0 E \oint dA = q_{enc}$.

$$\Rightarrow \epsilon_0 E (2\pi r h) = \lambda h.$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{line of charge})$$

This is the electric field due to an infinitely long, straight line of charge, at a point that is a radial distance r from the line. The direction of $\vec{E}$ is radially outward from the line of charge. The direction would be radially inward if it is negative.

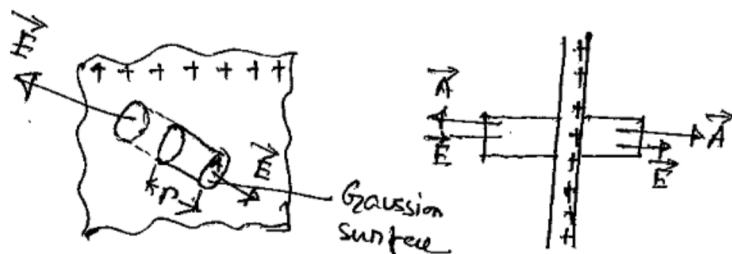
Planar Symmetry

Here,

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{enc} \quad \text{--- (i)}$$

$$\epsilon_0 (EA + EA) = \sigma A \quad \text{--- (ii)}$$

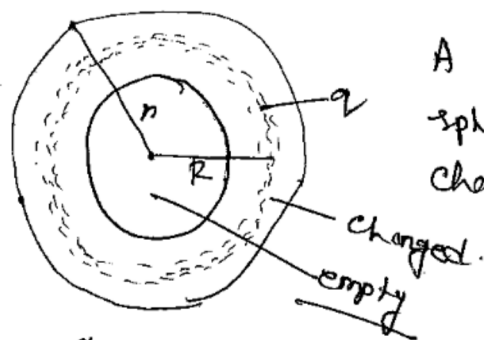
$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{sheet of charge}) \quad \sigma \rightarrow \text{Surface charge density.}$$



There is flux only through the two end faces.

Spherical Symmetry

A shell of uniform charge attracts or repels a charged particle that is outside the shell as if all the shell's charge were concentrated at the center of the shell.



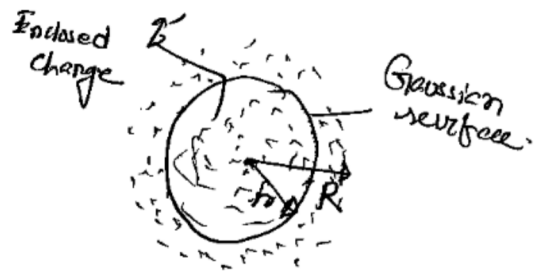
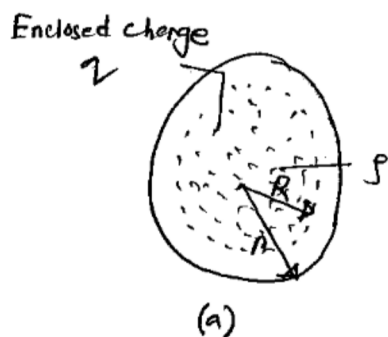
A thin, uniformly charged spherical shell with total charge q , in cross section.

at Gauss law, S_2 , $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ (spherical shell, field $r \geq R$)

Applying Gauss' law at S_1 , $r < R$, leads directly to $E = 0$.

$E = 0$ (Gaussian surface no charge). Thus if a charged particle were enclosed by the shell, the shell would exert no electrostatic force on the particle.

⊛ If a charged particle is located inside a shell of uniform charge, there is no electrostatic force on the particle from the shell.



3

$$E = \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2} \quad (\text{spherical distribution, field at } r \leq R)$$

If the full charge q enclosed within radius R is uniform, then q' enclosed within radius r is proportional to q .

$$\frac{\text{(Charge enclosed by sphere of radius } r)}{\text{(Volume enclosed by sphere of radius } r)}}{\text{(Volume enclosed by sphere of radius } R)}} = \frac{\text{full charge}}{\text{full volume}}$$

$$\text{or, } \frac{q'}{\frac{4}{3}\pi r^3} = \frac{q}{\frac{4}{3}\pi R^3}$$

$$q' = q \frac{r^3}{R^3}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q r^3}{R^3 r^2}$$

$$= \frac{q r}{4\pi\epsilon_0 R^3}$$

$$= \left(\frac{q}{4\pi\epsilon_0 R^3} \right) r \quad (\text{Uniform charge, field at } r \leq R)$$